CHETHAN KUMAR U

SDM Deployment Engineer

Nokia SDM | 5G UDM Deployment | Kubernetes/Helm | ATP & Service Validation

Indore, Madhya Pradesh, India | +91 8792155699 | kumarachetuchethan@gmail.com | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

SDM Deployment Engineer supporting Rakuten Mobile's cloud-native 5G Core and IMS Core with Nokia SDM/UDM deployment and integration activities. Hands-on with the deployment and pre-production validation of 5G UDM CNFs using Helm charts on Kubernetes, Nokia IMS HSS/SDL operations, and subscriber provisioning failure analysis via Nokia Flow One (B2B/B2C/Readlite). Experienced in Kubernetes/Helm/Docker operations, kubectl-based pod/log monitoring, Grafana and internal monitoring, Linux/Bash troubleshooting, 5G SBA networks (AMF, UDM, AUSF, NRF, SMF, SMSF, UPF), LTE/IMS/VoLTE/EPC operations, 5G registration/authentication/PDU session validation, alarm/KPI monitoring, RCA support, ITSM incident management and MOP/RMOP-driven changes. Seeking an SDM Deployment Engineer role to drive Nokia SDM/UDM deployment, integration, testing and service assurance across LTE/VoLTE/IMS/5G Core.

TECHNICAL SKILLS

Nokia SDM & 5G Subscriber Data Management: Nokia IMS HSS; SDL; SDM/HSS deployment & integration support; 5G UDM deployment support; UDM CNFs; AMF; AUSF; HSS/HLR; UDM; AUSF; SDL; EIR; OneNDS; PCRF; Nokia Flow One; B2B/B2C/Readlite provisioning workflows; Keycloak (user provisioning & secrets)

Deployment & Cloud-Native: Kubernetes; Docker containers; kubectl; Helm charts (configuration, installation, upgrade, rollback validation); Linux; Bash shell; pod access/logs/health checks; pre-production deployment; post-deployment verification; application/service/pod health validation; software upgrades & rollback support

Protocols & Analysis: SIP; Diameter; GTP; HTTP/2; S1AP; IMS/VoLTE call flows; 5G signaling analysis; packet captures (PCAP); Wireshark; CDR analysis; transaction logs; protocol-trace investigation

Telecom Architecture & Operations: IMS; EPC; VoLTE; 4G/5G Core; RAN/EPC/IMS/Core (Rakuten Mobile); LTE signaling; NSA/SA architecture; 5G SBA (AMF, UDM, AUSF, NRF, SMF, SMSF, UPF); subscriber management; LTE/VoLTE registration & mobility; Int'l Roaming

Monitoring, Testing & Support: ATP (Acceptance Test Procedures); MOP/RMOP execution; OSS monitoring; alarm/event monitoring; KPI/SLA monitoring; incident management; RCA support; service desk/ticket handling & analysis; vendor/L2/L3 coordination; Grafana; internal monitoring platforms

Tools & Programming: RADCOM; ForeSight; Wireshark; RCP; WinSCP; Xshell; PuTTY; MobaXterm; JIRA; Service Desk; Excel; Python; SQL; MySQL (basic); HTML; CSS

PROFESSIONAL EXPERIENCE

TECH MAHINDRA LTD. | **SDM Deployment Engineer (Nokia SDM / 5G UDM — Rakuten Mobile)** | Nov 2024 – Present

Client: Rakuten Mobile | Cloud-native 5G Core / IMS Deployment & Integration Support

- **5G UDM Deployment & Pre-Production Support:** Supported deployment and validation of the 5G User Data Management (UDM) using Helm charts on Kubernetes in Rakuten's cloud-native 5G Core. Assisted in Helm chart configuration, installation and upgrade; supported pre-production checks, configuration validation and basic post-deployment verification.
- **UDM Application Monitoring & Interworking Validation:** Monitored UDM application health using Kubernetes commands and logs. Supported validation of UDM interworking with AMF, AUSF, PCF and NRF, and validated 5G Registration, Authentication, PDU Session Establishment and Deregistration procedures.
- **Keycloak Provisioning & Secrets:** Created users and assigned groups in Keycloak for secrets provisioning and configuration, supporting secure subscriber-data and platform operations.
- **Nokia IMS HSS / SDL & Core Operations:** Supported production IMS Core operations for Rakuten Mobile across Nokia/Mavenir network functions in CNF/VNF environments, including HSS, SBC, DRA, STP, CFX (P-/S-CSCF), I-CSCF, PGW, SGW, PCRF, DNS, MRF, MGCF, BGCF, MSS and TAS.
- **Protocols & Signaling Diagnostics:** Performed end-to-end signaling analysis using Wireshark and PCAP traces; troubleshot IMS/VoLTE/5G call flows and service issues; analyzed SIP, Diameter, GTP, S1AP and HTTP/2 signaling, transaction logs and CDRs for fault isolation and subscriber-experience validation.
- **Subscriber Registration & Authentication Troubleshooting:** Troubleshot subscriber registration and authentication failures; supported alarm monitoring and structured Root Cause Analysis (RCA), contributing to faster incident investigation and resolution for customer-impacting issues.
- **Deployment Validation, Upgrades & Rollback:** Supported software upgrades, rollback validation and post-deployment health checks; executed Standard, Normal and Emergency Change Requests as per approved MOP/RMOP; supported pod-to-pod data migration with subsequent health, service and KPI checks.
- **Incident Management & Service Desk:** Handled service desk tickets and ITSM incidents through investigation, categorization, RCA support, escalation, status updates and closure coordination with vendor, L2/L3 and cross-functional teams; acted as shift lead when required.

EDUCATION

B.Tech, Electronics and Communication Engineering — Presidency University, Bengaluru | 2024 | CGPA: 8.29/10

ADDITIONAL INFORMATION

Languages: English, Kannada, Hindi, Telugu; basic Tamil

Portfolio: https://chethan-2001.github.io/portfolio-3d/